

TELECOM CUSTOMER CHURN DATASET ANALYSIS

Dashboard



1. Project Overview

The Telecom Customer Churn Analysis project focuses on understanding customer behavior and identifying the major factors responsible for customer churn in a telecommunications company.

The analysis was performed on the Telco Customer Churn dataset, which contains information about customer demographics, services, contract details, monthly charges, total charges, tenure, payment methods, and churn status.

The main purpose of this project is to convert raw customer data into useful business insights that can help the company improve customer retention, reduce churn, and make better business decisions.

2. Dataset Description

The dataset contains 7,043 customer records with 21 attributes.

Some of the important attributes analyzed include:

- **Customer information:** Gender, Senior Citizen, Partner, Dependents
- **Service information:** Phone Service, Internet Service, Online Security, Tech Support, Streaming Services
- **Contract information:** Contract Type, Paperless Billing, Payment Method
- **Financial information:** Monthly Charges and Total Charges
- **Customer relationship information:** Tenure
- **Target variable:** Churn

The Churn column indicates whether a customer has left the company (Yes) or is still using the service (No).

3. Data Preparation and Analysis

Before performing the analysis, the dataset was inspected and cleaned to make it suitable for analysis.

The main preprocessing steps included:

- Checking the dataset structure and data types.
- Converting Total Charges into a numerical format.
- Identifying and handling missing values.
- Checking for duplicate records.
- Converting churn status into a numerical form (0 and 1) for correlation analysis.
- Creating customer segments based on tenure.
- Categorizing customers based on their monthly spending.

After preprocessing, different statistical and visual analysis techniques were used to identify patterns in customer behavior.

4. Dashboard Analysis

The dashboard provides an interactive overview of customer retention and churn.

Key Performance Indicators

The dashboard shows four main KPIs:

- **Total Customers:** 7,043
- **Active Customers:** 5,174
- **Churned Customers:** 1,869
- **Churn Rate:** 26.54%

These KPIs provide a quick overview of the company's customer retention situation.

Visualizations

The dashboard contains the following major analyses:

- Churn by Contract Type
- Churn by Internet Service
- Monthly Charges Distribution
- Customer Tenure Analysis
- Correlation Heatmap

Filters for Gender, Contract Type, and Internet Service allow the data to be explored from different customer segments.

5. Business Insights

The analysis provided several important insights about customer churn.

1. Month-to-month customers have the highest churn

Customers with month-to-month contracts show considerably higher churn compared with customers on one-year and two-year contracts.

Business meaning: Customers without long-term commitments are more likely to leave the company.

2. Long-term contracts are associated with better retention

Customers with one-year and two-year contracts have much lower churn levels.

Business meaning: Long-term contracts help create customer stability and improve retention.

3. Fiber optic customers show comparatively high churn

Among internet service users, fiber optic customers have a relatively high number of churned customers.

Business meaning: The company should investigate whether pricing, service quality, technical issues, or customer expectations are contributing to churn in this segment.

4. New customers are more vulnerable to churn

The tenure analysis shows that customers with shorter tenure have a higher tendency to churn, while customers who have stayed longer are generally more stable.

Business meaning: The early customer experience is very important for retention.

5. Higher monthly charges can be associated with greater churn

The monthly charges analysis shows that churn is noticeable among customers with relatively higher monthly charges.

Business meaning: Customers paying more may have higher expectations and may be more sensitive to pricing or service quality.

6. Tenure and total charges have a strong relationship

The correlation analysis shows a strong positive relationship between tenure and total charges (approximately 0.83).

Business meaning: Customers who stay longer naturally generate more cumulative revenue for the company.

7. Monthly charges and total charges are positively related

There is also a positive relationship between monthly charges and total charges (approximately 0.65).

Business meaning: Higher-value customers can contribute significantly more revenue over their relationship with the company.

8. Churn is negatively related to tenure

The correlation between tenure and churn is approximately -0.35.

Business meaning: As customer tenure increases, the likelihood of churn generally decreases.

9. Payment method can be used as a retention signal

The analysis of customer payment behavior indicates that some payment methods are associated with comparatively higher churn.

Business meaning: Payment method can be considered as one of the factors when identifying customers who may require additional retention attention.

10. Customer retention has direct business value

Because long-tenure customers contribute more cumulative revenue, reducing churn—especially among new and high-value customers—can have a significant impact on the company's revenue.

Recommendations

Based on the analysis, the following business actions are recommended:

1. Promote Long-Term Contracts

Provide attractive discounts, additional benefits, or loyalty rewards to encourage month-to-month customers to move to one-year or two-year contracts.

2. Strengthen New Customer Onboarding

New customers should receive better onboarding, early support, service guidance, and follow-up during their first few months.

This can reduce early-stage churn.

3. Introduce Personalized Retention Offers

Customers with high monthly charges or higher churn risk can be offered personalized discounts, upgraded service packages, or additional benefits.

4. Improve Fiber Optic Customer Experience

Since fiber optic customers show comparatively high churn, the company should investigate:

- Service quality
- Network reliability
- Pricing
- Technical support

- Customer complaints

Targeted improvements can help reduce churn in this segment.

5. Encourage Loyalty

Long-term customers should receive loyalty benefits such as discounts, service upgrades, reward programs, or exclusive offers.

This can increase customer satisfaction and encourage customers to remain with the company.

6. Monitor High-Risk Customers

The company can create a customer churn-risk segment using factors such as:

- Short tenure
- Month-to-month contract
- High monthly charges
- Internet service type
- Payment method

These customers can be targeted with proactive retention campaigns.

Conclusion

The Telecom Customer Churn Analysis demonstrates how data analytics can help a telecom company understand why customers leave and which customer groups require attention.

The analysis shows that contract type, tenure, monthly charges, internet service, and payment behavior are important factors associated with customer churn.

The most important business opportunity is to focus on new customers and month-to-month customers, while also improving the experience of high-risk and high-value customers.

By using customer data for targeted retention strategies instead of applying the same approach to everyone, the company can potentially reduce churn, improve customer loyalty, and increase long-term revenue.